It is obtained by dividing the total land area by the total population, the quotient being the number of people per square kilometre. Compared with simple arithmetic density, physiological or nutritional density is a more refined method of calculating man-land ratios.

**Physiological or Nutritional** density is the ratio between total population and total cropped area. The total arable land in the world is **13.3%** and the nutritional density of the world is **325 per sq.km** of land. The total percentage of arable land is 48.83 in India and its nutritional density is 753 per sq.km of land. **Singapore** has the highest nutritional density of population of **440,998 per sq.km** of land the world. The areas of density of population can be divided into three as follows:

## 1. High density areas of population

Fertile plains with favourable climate and highly industrialised and urbanised areas are generally densely populated. There are four major areas of high density of population with **more than 100 persons** per sq.km. Areas include:

a. Eastern Asia, including china, Japan and Republic of Korea.

b. Southern Asia, comprising India, Bangladesh and Sri Lanka.

c. North-Eastern part of the United States of America.

d. Central and North-Western Europe.

Of the four regions given, the first two i.e. Eastern Asia and South Asia have high density of population due to favourable environmental conditions such as favourable climate, fertile soil and large areas of plains which encourage the growth of agriculture. The plains and river valleys of India and China are densely populated. In the last two groups i.e. North Eastern United States of America and North – Western Europe which are densely populated due to the concentration of manufacturing industries.

**HOTS**

Why has Singapore the highest physiological/nutritional density of population in the world?

## 2. Moderate density areas of population

The areas of moderate density of population have **between 10 and 80 persons** per sq.km. The areas of moderate density of population include Central part of the United States of America, Tropical Western Africa, Western blocks of Russia, Eastern Europe, Deccan Plateau of India, Central China, Southern portion of the Plateau of Mexico, North-Eastern Brazil and Central Chile, The above areas are characterised by the well developed agricultural activities, favourable climate, fertile soils, fishing, etc,.

## 3. Low density areas of population

About half the area of the world has population **less than 10 persons** per sq.km. Certain vast areas remain completely uninhabited. The main areas are

a. Amazon forest region of South America and Congo forest region of Africa.

b. Arctic area of Canada, Greenland and the Polar regions.

c. Great deserts of the world i.e. Sahara, Kalahari, Arabia, Great desert of Australia, Atacama Desert of South America, desert regions of Western United States and Thar Desert of India.

d. High mountainous regions in all continents.

e. Antarctica.



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